Mahesh Jivaji Shriram

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Professional Summary

AI Engineer with experience building production-grade **LLM-powered applications, Agentic AI systems, and retrieval-augmented solutions**. Experienced in designing **multi-agent workflows, AI-assisted engineering systems, MCP integrations and AI automation pipelines** for enterprise software engineering use cases. Hands-on exposure to **LLM fine-tuning, knowledge distillation, model quantization, embeddings, semantic search, and model evaluation**, with a strong foundation in **Machine Learning, NLP, PyTorch, TensorFlow and Python**. Proven experience taking AI solutions from experimentation to production with measurable engineering impact.

Technical Skills

- **Programming Languages:** Python, C++, JavaScript
- **ML/DL:** Scikit-learn, TensorFlow, PyTorch, NLP, Feature Engineering, Hyperparameter Tuning, Model Evaluation
- **LLM Engineering:** Hugging Face, Transformers, LLM Fine-Tuning, SFT, LoRA/QLoRA, Knowledge Distillation, Model Quantization, Embeddings, RAG
- **Context Engineering:** Context Construction, Prompt Engineering, Context Management, Retrieval Optimization, Metadata Enrichment, Structured Outputs
- **Agentic AI:** LangChain, LangGraph, Multi-Agent Systems, MCP, Agentic Workflows, Tool Calling, Human-in-the-Loop (HITL), Guardrails
- **Evaluation & Experimentation:** RAGAS, DeepEval, MLflow, Weights & Biases, LLM Evaluation, Benchmarking, Experiment Tracking, Regression Testing
- **Backend, Data & Deployment:** FastAPI, REST APIs, PostgreSQL, pgvector, MySQL, Firebase, Vector Databases, GitHub Actions, CI/CD, Azure App Service, Azure Static Web Apps

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Experience

Harbinger Group — July 2025 – Present

AI Engineer

Adaptive Learning Intelligence Platform - Personalized AI Learning

- Built an **AI-powered learning platform** that generates personalized learning packages based on learner profiles, learning objectives, and relevant course content.
- Developed a **custom document ingestion pipeline** for content extraction, chunking, embedding generation, metadata enrichment, and vector indexing.
- Engineered **context-aware hierarchical retrieval** using parent-child relationships and related learning concepts to preserve context and identify relevant course content.
- Deployed the React frontend on **Azure Static Web Apps** and backend on **Azure App Service**, integrating Azure vector search and PostgreSQL for retrieval and metadata management.

Bug-Fixing Agent – AI-Powered Defect Resolution System

- Built a **multi-agent system** for defect analysis, root-cause identification, fix generation, and validation across **.NET and Angular**.
- Automated the **Jira-to-PR workflow** using **GitHub Agentic Workflows and MCP**, integrating

Atlassian, GitHub, and CodeGraph for code analysis and autonomous fix generation without human intervention.
- Achieved **84% success across 42 defects**, saving **207+ engineering hours**, with guardrails and browser-assisted validation.

AI-Assisted Engineering Initiative

- Built **AI-assisted engineering workflows** for code generation, auditing, and validation using **LLMs, MCP, GitHub, AI Hooks, and CI/CD** across enterprise teams.
- Implemented **spec-driven workflows and pre/post-tool guardrails** to improve the reliability and consistency of AI-generated changes.
- Saved **1,875+ engineering hours across 238 tasks**, achieving **90.8% backend** and **88.6% frontend accuracy**.

Hartaraf Nov 2024 – Feb 2025
Software Engineer Intern (Remote)

- Developed an **intent classification and NLP pipeline** for natural-language service discovery on a Flutter-based platform.
- Implemented **Speech-to-Text (STT) and Text-to-Speech (TTS)** pipelines for voice-based interactions.
- Integrated **RAG-based service matching** with Firebase and REST APIs to retrieve contextually relevant local service providers.

Independent Projects

ModelForge - LLM Fine-Tuning, Distillation & Quantization

- Built a **model adaptation pipeline** for experimenting with **SFT, LoRA/QLoRA, knowledge distillation, and model quantization** on compact instruction-tuned LLMs.
- Implemented **response and logit distillation** to transfer capabilities from larger models to smaller models while preserving task performance.
- Compared **base, fine-tuned, distilled, and quantized models** across quality, inference latency, and memory footprint.
- Analyzed the trade-offs between **model quality, efficiency, and resource requirements** for deploying LLMs in resource-constrained environments.

ModelLens - LLM Evaluation & Benchmarking Framework

- Built an **automated evaluation framework** using **RAGAS and DeepEval** to measure faithfulness, answer relevance, context precision, and hallucination-related failures.
- Designed **domain-specific evaluation datasets and benchmarking workflows** to compare base, fine-tuned, and quantized models across quality and inference performance.
- Integrated **Weights & Biases (W&B)** for experiment tracking, metric analysis, and regression detection across model iterations.

Education

Bachelor of Engineering in Artificial Intelligence and Data Science
PES Modern College of Engineering, Pune August 2021 – June 2025
Cumulative GPA: 8.2